

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Summer Semester Examination – 2026

Course: B. Tech. Branch : ELECTRICAL ENGG. Semester : IV

Subject Code & Name: Electrical Machine-I 25AF1293PC402

Max Marks: 60

Date:13/07/2026

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 Which of the following transformer losses remains almost constant from no-load to full-load? A. Copper loss B. Eddy current loss C. Iron (Core) loss D. Stray load loss		1
2 Which transformer has only one continuous winding? A. Distribution transformer B. Instrument transformer C. Auto transformer D. Welding transformer		1
3 Which transformer connection provides a neutral point? A. Δ - Δ B. Δ -Y C. Y-Y D. Both B and C		1
4 Open-delta connection can supply approximately what percentage of the original three-phase transformer bank rating? A. 100% B. 86.6% C. 57.7% D. 66.7%		1
5 Which factor affects the selection of transformer connection? A. Harmonics B. Neutral requirement C. Load type D. All of the above		1
6 Scott connection is primarily used for: A. Three-phase to single-phase conversion B. Single-phase to three-phase conversion C. Three-phase to two-phase conversion D. DC to AC conversion		1

- 7 The purpose of an On-Load Tap Changer (OLTC) is to: 1
 A. Increase frequency
 B. Control voltage without interrupting supply
 C. Reduce transformer losses
 D. Improve power factor
- 8 The operating principle of a DC generator is based on: 1
 A. Electrostatic induction
 B. Electromagnetic induction
 C. Mutual capacitance
 D. Self-induction only
- 9 Armature reaction mainly causes: 1
 A. Increased insulation resistance
 B. Distortion and weakening of the main field flux
 C. Reduction in copper loss
 D. Increase in mechanical strength
- 10 The Open Circuit Characteristic (OCC) of a DC generator is the graph between: 1
 A. Torque and Speed
 B. Generated EMF and Field Current
 C. Voltage and Load Current
 D. Speed and Current
- 11 Which starter is commonly used for DC shunt motors? 1
 A. Star-Delta Starter B. DOL Starter
 C. Three-point Starter D. Auto-transformer Starter
- 12 In Hopkinson's test, two identical DC machines are connected: 1
 A. In series
 B. Mechanically coupled and electrically connected in parallel
 C. In delta connection
 D. In star connection
- Q. 2 Solve the following.** 12
- A) Explain the working principle of a single-phase transformer. 6
- B) Explain an ideal transformer with its phasor diagram. 6
- Q.3 Solve the following.** 12
- A) What is braking? Explain dynamic braking with a labelled diagram in DC machine.. 6

- B) Explain the construction of a DC machine with a labelled diagram.**

6

Q. 4 Solve Any Two of the following.

12

- A) Derive the EMF equation of a single-phase transformer.**
- B) A DC generator has an armature EMF of 100 V when the useful flux per pole is 20 mWb and the speed is 800 rpm. Calculate the generated EMF:**
- i) When the flux remains the same and the speed is 1000 rpm**
 - ii) When the flux per pole is 24 mWb and the speed is 900 rpm.**
- C) Explain the welding transformer in detail.**

6

6

6

Q.5 Solve Any Two of the following.

12

- A) Derive the torque equation of a DC machine**
- B) What is the necessity of parallel operation? Explain the parallel operation of transformers**
- C) A 230/110 V, single-phase transformer takes an input of 350 VA at no-load and rated voltage. The core loss is 110 W. Determine:**
- i) Iron-loss component of no-load current**
 - ii) Magnetizing component of no-load current**
 - iii) No-load power factor.** <https://www.batuonline.com>

6

6

6

Q. 6 Solve Any Two of the following.

12

- A) Explain the armature voltage control method.**
- B) Explain the performance characteristics of a three-phase transformer.**
- C) Explain voltage build-up in a self-excited DC generator and Scott connection in detail.**

6

6

6

***** End *****